

Saturated Superfluid Vacuum VII-a

Quantum Mechanics from Hydrodynamics

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Abstract

This paper derives the quantum-mechanical limit of the Saturated Superfluid Vacuum (SSV) framework. The scope is deliberately narrow: the Madelung transformation, the linear Schrödinger limit, uncertainty as a hydraulic localisation bound, and measurement as a physical reconnection channel.

The formal result is the recovery of the Schrödinger equation from the low-amplitude hydrodynamic regime of the SSV medium. Quantisation enters as the single-valuedness condition on the order-parameter phase, tying discrete winding numbers to circulation topology rather than to a separate quantisation rule. Uncertainty is interpreted as the hydraulic cost of localising the order parameter. Measurement is modelled as an irreversible reconnection or wake-writing event in the medium.

Claims about emergent geometry, Einstein equations, cosmogony, and full-series synthesis are left to companion papers.

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1 Introduction

The SSV programme treats the vacuum as a saturated superfluid described by a complex order parameter Ψ . Papers I–IV establish the medium, its topological defects, thermodynamic arrow, and gravitational limit. Paper V extends the same picture to strong gravity and condensate black holes. The present paper asks a narrower question: under what conditions does ordinary quantum mechanics appear as the low-amplitude hydrodynamic limit of this medium?

The working thesis is that quantum behaviour is not an additional foundational axiom. It is the wave-mechanical regime of the order parameter when perturbations are small, gradients are gentle, and defects can be treated through an effective potential. In that regime the Madelung variables map the complex wave equation to continuity plus Euler-type flow equations, and the inverse map recovers the Schrödinger equation for a slowly varying envelope.

This paper separates three claim levels explicitly.

- C1. Formal derivation.** The Madelung transformation and its inverse show how a Schrödinger-type equation is equivalent to hydrodynamic flow with a quantum-pressure term. This is an exact algebraic equivalence, not an approximation.
- C2. Controlled limit.** In the low-amplitude, long-wavelength regime, the nonlinear medium terms can be absorbed into background phase and effective potentials, leaving the ordinary linear Schrödinger equation. This is the limit that makes SSV contact with standard quantum mechanics.
- C3. Physical interpretation.** Quantisation, uncertainty, and measurement are read as circulation topology, localisation stiffness, and irreversible reconnection respectively. These are model commitments that arise naturally from the medium picture; they are not independent mathematical theorems.

Roadmap

Section 2 carries out the Madelung transformation and derives the hydrodynamic equations. Section 3 takes the Schrödinger limit and shows how quantisation follows from phase topology. Section 4 interprets the uncertainty relation as a hydraulic localisation bound. Section 5 models measurement as a reconnection event and discusses Born weights. Section 6 places the results in the context of the series and lists open calculations.

2 Madelung Transformation

2.1 Polar decomposition

Let the local state of the saturated medium be the complex order parameter

$$\Psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{iS(\mathbf{x}, t)/\hbar}, \quad (1)$$

where ρ is the mass density and S is the phase action. The associated irrotational velocity field is

$$\mathbf{v} = \frac{1}{m} \nabla S. \quad (2)$$

For the logarithmic-Schrödinger (LogSE) medium of Papers I–IV [1, 4], the order parameter obeys

$$i\hbar \partial_t \Psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}} + U(\rho) \right] \Psi, \quad (3)$$

where V_{ext} is a slowly varying effective potential and $U(\rho)$ is the medium self-interaction. The specific equation of state is fixed in earlier papers; the present paper uses only the hydrodynamic structure common to this class of order-parameter equations.

2.2 Hydrodynamic form

Substituting (1) into (3) and separating real and imaginary parts gives the Madelung pair

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (4)$$

$$\partial_t S + \frac{(\nabla S)^2}{2m} + V_{\text{ext}} + U(\rho) + Q = 0, \quad (5)$$

with the quantum pressure

$$Q(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \quad (6)$$

Taking a gradient of (5) gives the Euler equation

$$m(\partial_t + \mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla (V_{\text{ext}} + U(\rho) + Q). \quad (7)$$

Madelung equivalence

Equations (4)–(7) are the Madelung equations: compressible irrotational hydrodynamics with an additional quantum-pressure contribution Q . The transformation is exact and bidirectional: starting from the wave equation (3) yields the hydrodynamic pair; starting from the hydrodynamic pair with the irrotationality condition (2) reconstructs the wave equation.

3 Schrödinger Limit

3.1 Low-amplitude long-wavelength limit

Write the density as $\rho = \rho_0 + \delta\rho$, with $|\delta\rho| \ll \rho_0$, and assume gradients are long compared with the healing length of the medium. Expand the self-interaction around the saturated background:

$$U(\rho) = U(\rho_0) + U'(\rho_0) \delta\rho + \dots \quad (8)$$

The constant term shifts the global phase; the linear term is either negligible for a sufficiently weak perturbation or is absorbed into an effective potential. The slowly varying envelope ψ then obeys

$$i\hbar \partial_t \psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}} \right] \psi, \quad (9)$$

the ordinary Schrödinger equation for the effective degree of freedom.

Schrödinger limit

The SSV medium in its low-amplitude, long-wavelength regime reduces exactly to linear Schrödinger dynamics. The claim is not that all of quantum theory is derived in one step; it is that whenever the conditions above hold, the same hydrodynamic variables reduce to the linear dynamics used in standard quantum mechanics.

Deferred: bound-state spectra from LogSE

Equation (9) recovers the Schrödinger form, but the effective potential V_{eff} must be computed from the full LogSE background for each physical situation. Deriving the hydrogen-like bound-state spectrum directly from the LogSE—without invoking V_{eff} as an external input—is part of the Paper I numerical programme [1].

3.2 Quantisation as circulation topology

The phase in (1) must be single-valued around any closed loop that does not cross a zero of Ψ . Therefore

$$\oint_C \nabla S \cdot d\mathbf{l} = 2\pi n\hbar, \quad n \in \mathbb{Z}. \quad (10)$$

Using (2), this is the circulation condition

$$\oint_C \mathbf{v} \cdot d\mathbf{l} = \frac{2\pi n\hbar}{m}. \quad (11)$$

Quantisation is therefore tied to the topology of the phase field. It is not inserted as a separate rule for allowed orbits; it is the consequence of a single-valued order parameter with vortex cores acting as excluded phase singularities. The detailed spectrum of any bound system depends on its effective potential and boundary conditions, but the origin of discrete winding numbers is hydrodynamic. This is consistent with the topological spin derivation of Paper V [5] and the vortex-defect picture of Paper I [1].

4 Uncertainty as Hydraulic Bound

The standard uncertainty relation follows from Fourier analysis of wave packets:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (12)$$

SSV adds a physical interpretation of the same relation. A perturbation localised to a short length scale must include large gradients of $\sqrt{\rho}$ and phase. Through (6), those gradients carry a quantum-pressure cost $\Delta E_Q \sim \hbar^2/(2m \Delta x^2)$. Pushing Δx downward therefore forces a larger spread in momentum modes and a larger hydraulic stiffness penalty, arriving at (12).

This section does not replace the standard proof. The proof remains the wave-packet Fourier theorem. The SSV claim is that the theorem has a medium-level interpretation: localisation is limited because the order parameter cannot be squeezed arbitrarily without exciting the pressure and phase-gradient channels of the fluid. The uncertainty principle is the hydraulic stiffness of the vacuum medium made quantitative.

Deferred: precise $\hbar/2$ prefactor from LogSE

The hydraulic-stiffness argument recovers the form of (12) and the $\hbar$ scaling, but deriving the precise prefactor $1/2$ from the LogSE equation of state—without importing it from the standard wave-packet calculation—requires computing the minimum uncertainty product for a LogSE coherent state. This is a tractable calculation deferred to the Paper I programme [1].

5 Measurement as Reconnection

5.1 Irreversibility from topology change

The measurement argument is kept narrow. A measurement is a physical interaction that correlates a microscopic degree of freedom with a macroscopic apparatus by driving some part of the combined medium into an irreversible reconnection or wake-writing event.

Before measurement, a superposition is represented by a coherent field configuration,

$$\Psi = \sum_k c_k \psi_k. \quad (13)$$

As long as the relevant branches remain phase-coherent, interference terms are available. A measurement interaction is the regime in which the probe, apparatus, and surrounding medium cross a threshold where topology changes, phonons are emitted, or persistent wake degrees of freedom are written—the same irreversibility mechanism described for time in Paper III [3]. Reversing the event would require restoring the original phase relation across all of those channels, not merely evolving a small isolated wave packet backward.

5.2 Born weights as a working model

The natural branch weights in (13) are $|c_k|^2$ because the local energy and intensity available to drive an apparatus response are quadratic in the wave amplitude. This gives the usual Born form

$$P_k = |c_k|^2. \quad (14)$$

At the present stage this is a consistency argument, not a full derivation.

Deferred: Born-rule basin calculation

The missing calculation is a reconnection-threshold model showing that the basin volume for outcome k is proportional to the incoming branch intensity $|c_k|^2$ under realistic apparatus coupling. This would promote the Born-rule argument from physical interpretation to formal derivation. It requires a reduced model of a two-channel apparatus coupled to the SSV medium and is deferred to a companion analysis.

5.3 What this does and does not claim

This picture does not require consciousness, an external classical realm, or a new postulate of collapse. It also does not yet derive a complete theory of detector statistics. Its useful claim is more modest: SSV supplies a candidate physical channel for irreversibility during measurement, namely topology change and wake formation in the medium. This is consistent with, but not identical to, decoherence programmes that source irreversibility from environmental entanglement; the SSV version sources it from physical medium events.

6 Discussion

6.1 Position in the series

Paper VII-a is the QM-limit paper in the SSV series. Its inputs are the Madelung structure of the LogSE (Papers I–IV) and the topological defect picture (Paper I, Paper V). Its output—the Schrödinger limit, circulation quantisation, hydraulic uncertainty, and reconnection measurement—is passed forward as a standing result. Claims about emergent geometry and the Einstein-equation limit are in Paper VII-b; cosmogony is in Paper VIII.

6.2 Relation to the Madelung and pilot-wave literature

The Madelung [7] representation is well-established; this is not a new mathematical result. The SSV contribution is the physical identification of the medium: the Madelung fluid is not an abstract probability fluid but the actual vacuum condensate, with the same medium that supports topological defects, irreversible wakes, and gravitational gradients. The difference from the de Broglie–Bohm pilot-wave interpretation is that SSV does not introduce a guiding wave on top of classical trajectories; the order parameter IS the physical medium in which all dynamics occurs.

6.3 Testable consequences

Two near-term checks follow from the reconnection model.

- Q1. Reconnection signatures in controlled measurement systems.** If detector clicks are accompanied by irreversible medium events, then sufficiently sensitive optomechanical or superconducting measurement platforms should show excess correlated acoustic, thermal, or phase-noise signatures at the transition from reversible coupling to recorded outcome.
- Q2. Born-rule basin scaling.** A reduced reconnection-threshold model should test whether branch basin volumes scale as $|c_k|^2$ for simple two-channel apparatus couplings. This is the calculation needed before the measurement section can be promoted from physical interpretation to formal derivation.

6.4 Open problems

1. *Bound-state spectra.* Deriving the hydrogen-like spectrum directly from the LogSE, without treating V_{eff} as an external input, is part of the Paper I programme.
2. *Precise uncertainty prefactor.* The $1/2$ in (12) should emerge from the LogSE coherent-state minimum; the calculation is deferred to the Paper I programme.
3. *Multi-particle states.* The present treatment addresses single-order-parameter modes. Multi-particle states and exchange statistics require a field-theoretic extension of the Madelung picture, not developed here.
4. *Spin-1/2 and Fermi statistics.* The order parameter Ψ is a scalar; half-integer spin and fermion anti-symmetry require an extension to spinor order parameters or a separate topological argument. This is noted as a gap, not filled here.

7 Conclusion

SSV VII-a isolates the quantum-mechanics limit from the geometry and cosmogony claims elsewhere in the series. The paper's formal core is the Madelung bridge and the Schrödinger limit. Its interpretive core has three parts:

1. Quantisation reflects phase topology: discrete winding numbers follow from the single-valuedness of the order parameter, not from a separate quantisation axiom.
2. Uncertainty reflects hydraulic localisation stiffness: the quantum pressure (6) penalises tight localisation, recovering the Heisenberg bound from the equation of state.
3. Measurement reflects irreversible reconnection or wake-writing in the medium, sourcing irreversibility from topology change rather than from an external classical realm.

That is sufficient to make the paper standalone without requiring it to carry the Einstein-equation or cosmogony arguments.

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